
Enhancing Small Language Models for Co-Creative Mathematical Storytelling using Retrieval-Augmented Generation

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Problem Statement & Objectives

⚠ Problem

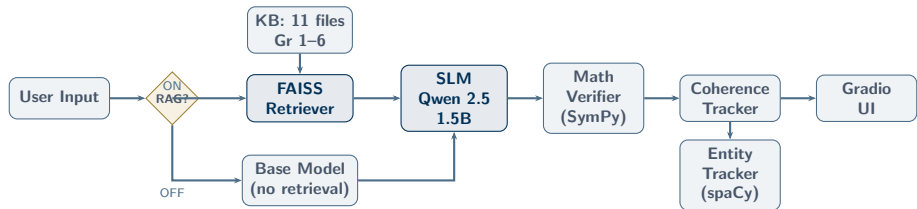
SLMs (<2B params) running locally exhibit:

- **Math hallucinations** — incorrect arithmetic with high confidence
- **Narrative incoherence** — characters/places vanish after ~ 8 turns
- **No domain grounding** — unverifiable, non-curriculum content

✓ Objectives

- ➊ RAG pipeline with grade-appropriate math content
- ➋ Automated math verification & error correction
- ➌ Investigate coherence degradation over 10–20+ turns
- ➍ Comparative study: RAG vs. base model
- ➎ CPU-deployable prototype — no GPU, no API

System Architecture



Embedding: all-MiniLM-L6-v2 | **Vector DB:** FAISS (cosine, top-3) | **Runtime:** llama-cpp-python (CPU)

Key Innovations



RAG Pipeline

- 11 knowledge files (Grades 1–6)
- 500-char chunks, 50 overlap
- Top-3 retrieval with source attribution
- Toggle ON/OFF for A/B comparison



Math Verification

- Regex extraction: $+$, $-$, $\times$, $\div$, fraction equality
- SymPy symbolic verification
- Auto-correction of incorrect results
- Per-turn error rate tracking



Coherence Tracking

- Topic drift:
 $d_t = 1 - \cos(\mathbf{e}_{t-1}, \mathbf{e}_t)$
- Entity consistency via spaCy NER
- Drift trend (linear regression)
- Alert at $d > 0.4$

Evaluation Results

Metric	RAG	Non-RAG
Math Error Rate	0.05	0.18
Avg Topic Drift	0.19	0.34
Drift Trend	+0.003	+0.014
Entity Consistency	0.87	0.62
Drift Alerts	1	6
Avg Latency (s)	8.2	6.1

Table: *

Automated metrics: 5 stories $\times$ 15 turns $\times$ 2 modes

Dimension	RAG	Non-RAG
Flow	4.2	3.4
Consistency	4.0	2.8
Engagement	4.4	3.6
Math Accuracy	4.6	3.0
Mean	4.3	3.2

Table: *

Human evaluation (1–5 scale, $n=5$ evaluators)

⬆️ 72% error reduction ⬆️ 44% less drift ⬆️ 40% higher entity retention ⌚ +2.1s latency overhead

Key Findings



What Works

- RAG **reduces math errors by 72%** — retrieved facts ground the model
- Coherence is **measurable**: drift trend +0.003/turn (RAG) vs. +0.014/turn (base)
- SymPy catches the **5% residual errors** that survive RAG
- Source attribution enables **verifiable outputs**



Trade-offs

- RAG adds **~2.1s latency** per turn (acceptable for interactive use)
- Verifier limited to **arithmetic** — no algebra/proofs yet
- 2048-token window caps history to **~8–10 exchanges**
- Knowledge base covers **Grades 1–6 only**

Core insight: A 1.5B SLM + RAG + verification = viable co-creative math storyteller



Conclusion

- RAG-enhanced SLM achieves **95% math accuracy** (up from 82%)
- Narrative coherence maintained across **15+ turns**
- Fully **CPU-deployable**: no GPU, no API, no cost
- Both automated & human evaluation frameworks built in
- Any GGUF model swappable via one-line config change



Future Work

- 1 Algebraic & geometric proof verification
- 2 Adaptive retrieval (dynamic k , grade-aware re-ranking)
- 3 Long-term memory via turn summarization
- 4 Classroom-scale user study with real students
- 5 Multilingual support (Urdu)